

Linear Equations with Distribution and Fractions

Answer Key

Grade 8 / Pre-Algebra — Expressions and Equations

Quick Answers

1. 3	4. -6	7. 8	10. 12
2. -2	5. $3x + 8$	8. 5	11. 9
3. 10	6. 14	9. $22 - 6x$	12. 7

Each solution shows the same four moves in the same order: clear fractions, distribute, collect, isolate. Where a step is the one students most often get wrong, it is called out. Substituting the answer back into the *original* equation is the check to insist on.

1. Solve $3(x + 4) = 21$.

Solution: Distribute the 3 across both terms in the brackets, then undo.

$$3x + 12 = 21$$

$$3x = 9$$

$$x = 3$$

$x = 3$

Check: $3(3 + 4) = 3 \cdot 7 = 21$. *Faster alternative:* divide both sides by 3 first, giving $x + 4 = 7$.

2. Solve $-2(x - 5) = 14$.

Solution: The factor is -2 , so both products change sign.

$$-2x + 10 = 14$$

$$-2x = 4$$

$$x = -2$$

$x = -2$

Common error: writing $-2x - 10 = 14$ (sign not distributed to the -5) gives $x = -12$.

3. Solve $\frac{x}{3} + \frac{x}{6} = 5$.

Solution: Denominators 3 and 6: the LCD is 6. Multiply *every* term by 6.

$$6 \cdot \frac{x}{3} + 6 \cdot \frac{x}{6} = 6 \cdot 5$$

$$2x + x = 30$$

$$3x = 30$$

$$x = 10$$

$$x = 10$$

Check: $\frac{10}{3} + \frac{10}{6} = \frac{20}{6} + \frac{10}{6} = 5$.

4. Solve $4 - 2(x + 3) = 10$.

Solution: The 2 being subtracted distributes as -2 ; only then combine the constants.

$$4 - 2x - 6 = 10$$

$$-2x - 2 = 10$$

$$-2x = 12$$

$$x = -6$$

$$x = -6$$

Common error: subtracting first ($2 \cdot (x + 3) = 10$, i.e. treating $4 - 2$ as 2) gives $x = 2$ — distribution comes before that subtraction.

5. Multiply $\frac{x}{4} + \frac{2}{3}$ by the LCD, 12, and write it without fractions.

Solution: Multiply each term separately: $12 \cdot \frac{x}{4} = 3x$ and $12 \cdot \frac{2}{3} = 8$.

$$3x + 8$$

Why it works: 12 is a multiple of both 4 and 3, so each denominator divides out exactly. Multiplying only the first term is the usual slip.

6. Solve $3(x - 2) = 2(x + 4)$.

Solution: Distribute on both sides, then collect the variable on the left.

$$3x - 6 = 2x + 8$$

$$x - 6 = 8$$

$$x = 14$$

$$x = 14$$

Check: $3(14 - 2) = 36$ and $2(14 + 4) = 36$.

7. Solve $\frac{3x}{4} - 2 = \frac{x}{2}$.

Solution: LCD of 4 and 2 is 4; multiply every term, including the -2 .

$$3x - 8 = 2x$$

$$x - 8 = 0$$

$$x = 8$$

$$x = 8$$

Common error: forgetting to multiply the whole-number term gives $3x - 2 = 2x$ and $x = 2$.

8. Solve $\frac{1}{2}(6x - 8) = 11$.

Solution: A fraction in front of brackets still distributes: halve each term.

$$3x - 4 = 11$$

$$3x = 15$$

$$x = 5$$

$$x = 5$$

Equivalent route: multiply both sides by 2 first, giving $6x - 8 = 22$.

9. Expand and simplify $7 - 3(2x - 5)$.

Solution: Distribute -3 over both terms, then combine the constants.

$$7 - 6x + 15 = 22 - 6x$$

$$22 - 6x$$

Common error: $7 - 6x - 15 = -8 - 6x$ — the second sign was not changed.

10. Solve $\frac{2}{3}(x + 6) = \frac{1}{2}(x + 2) + 5$.

Solution: LCD of 3 and 2 is 6. Multiply every term by 6 (keeping the brackets), then distribute.

$$4(x + 6) = 3(x + 2) + 30$$

$$4x + 24 = 3x + 6 + 30$$

$$4x + 24 = 3x + 36$$

$$x = 12$$

$$x = 12$$

Check: $\frac{2}{3}(18) = 12$ and $\frac{1}{2}(14) + 5 = 12$.

11. Solve $\frac{x + 3}{2} - \frac{x - 5}{4} = 5$.

Solution: The bar groups the whole numerator, so keep brackets when multiplying by the LCD 4.

$$2(x + 3) - (x - 5) = 20$$

$$2x + 6 - x + 5 = 20$$

$$x + 11 = 20$$

$$x = 9$$

$$x = 9$$

Common error: $2x + 6 - x - 5$ — the minus in front of the second fraction must reach both terms of its numerator; that slip gives $x = 19$.

12. Challenge: solve $\frac{2}{5}(10x - 15) = 3(x - 1) + 4$.

Solution: $\frac{2}{5}$ distributes over both terms; $\frac{2}{5} \cdot 10x = 4x$ and $\frac{2}{5} \cdot 15 = 6$.

$$4x - 6 = 3x - 3 + 4$$

$$4x - 6 = 3x + 1$$

$$x = 7$$

$$x = 7$$

Check in the original: $\frac{2}{5}(70 - 15) = \frac{2}{5}(55) = 22$, and $3(7 - 1) + 4 = 18 + 4 = 22$. Both sides agree, so $x = 7$ is confirmed.